

03_enzyme_oscillations

June 15, 2026

1 Biochemical Oscillations

1.1 1. Rhythms inside cells

Many processes in living cells are not static — they pulse, cycle and oscillate. Cytosolic calcium spikes every few seconds during hormone signalling, glycolytic intermediates in yeast oscillate with a period of minutes, the cell-division cycle is itself a slow relaxation oscillator, and circadian clocks tick with a roughly 24 h period. None of these rhythms need an external pacemaker: they are generated **autonomously** by the chemical kinetics of the reaction network itself. A network that “ticks” on its own, with nothing outside it providing the beat, is a striking example of **self-organisation** in biology — temporal order emerging spontaneously from molecular interactions.

This notebook works through a minimal example of such a system: a two-variable enzyme model with *forward* and *feedback* inhibition. It is small enough to simulate, plot and analyse with standard Python (`numpy`, `scipy`, `matplotlib`, `plotly`), yet rich enough to show the core ingredients that make biochemical oscillators tick.

1.2 2. The recipe: feedback + delay

A simple negative feedback loop — “if X gets too high, reduce its production” — tends to settle smoothly to a steady state, the way a thermostat keeps a room at a fixed temperature. Oscillations appear when a **delay** is added to that feedback: the correction arrives too late, the system overshoots, the feedback then over-corrects in the other direction, and the cycle repeats indefinitely.

In molecular terms, delay can come from a chain of reaction steps, from a build-up to some threshold before an inhibitory step switches on, or from cooperative (“ultrasensitive”) kinetics that make the response highly nonlinear. Tyson, Chen and Novak (2003) call a delayed negative-feedback loop a “**blinker**” — one of a small number of recurring dynamical motifs (*sniffers*, *buzzers*, *toggle switches* and *blinkers*) built from different combinations of positive and negative feedback in cell signalling networks.

1.3 3. Limit cycles and relaxation oscillators

Mathematically, a sustained oscillation corresponds to a **limit cycle**: a closed loop in state space (here, the S-P plane) that nearby trajectories spiral towards, regardless of where they start. This is qualitatively different from a system that simply rings briefly and then settles down — a limit cycle is approached *and then followed forever*.

Many biological oscillators are **relaxation oscillators**: one variable changes slowly while it “charges up”, then a second variable changes very rapidly once a threshold is crossed, “discharging”

the system before the slow build-up begins again. This produces the characteristic spike-like or sawtooth-like time courses seen in calcium spikes, action potentials, and the cell cycle.

1.4 4. Why this matters in cell physiology

Oscillations are not a curiosity — cells use them to encode and process information. A classic example is intracellular calcium signalling: Woods, Cuthbertson and Cobbold (1986) showed that hormone-stimulated liver cells do not raise their calcium concentration smoothly, but instead fire **repetitive transient spikes**, with the *frequency* of spiking (rather than its amplitude) encoding the strength of the hormonal signal. Similar oscillatory strategies are used in glycolysis, in Ca²⁺-induced Ca²⁺ release, in the cell-division cycle (cyclin-CDK oscillations), and in circadian clocks. Understanding the simple dynamical mechanisms behind these rhythms — feedback, delay, nonlinearity — is therefore directly relevant to understanding how cells sense, decide and time their behaviour (Tyson, Chen & Novak, 2003; Fall et al., 2002).

1.5 5. What this notebook does

1. **Time series** — simulate the model for two values of the parameter `a1` and compare a steady state to an oscillation.
2. **State-space nullclines** — visualise *why* the system behaves the way it does, using nullclines, fixed points and the limit cycle.
3. **Oscillation animation** — an intuitive picture of S and P oscillating out of phase.
4. **Bifurcation analysis** — using the companion script `enzym_oscillations_bifurcation.py`, systematically scan `a1` to map out exactly where oscillations occur.
5. **Summary and further reading** — what the simulations tell us, and where to learn more.

```
[1]: from scipy.integrate import solve_ivp
from matplotlib.pyplot import subplots
import numpy as np
```

2 Forward inhibition combined with feedback inhibition

2.1 The model

The model tracks two state variables, a “substrate” S and a “product” P — think of them as the concentrations of two interconverting metabolites:

$$\begin{aligned}\frac{dS}{dt} &= a_1 - b_1 S - v(S, P) \\ \frac{dP}{dt} &= a_2 - b_2 P + v(S, P)\end{aligned}$$

with the enzymatic conversion rate

$$v(S, P) = \frac{k_{max} S^n}{K_m^m + S^m} \cdot \frac{1}{1 + k_i P^q}$$

Reading the equations biologically:

- a_1 and a_2 — constant **production** (“inflow”) rates of S and P .
- b_1 and b_2 — first-order **removal** (“outflow”) rates, e.g. dilution or degradation.
- $v(S, P)$ — the rate at which an enzyme converts S into P . The same flux leaves S and arrives in P , so this term appears with opposite signs in the two equations.
 - $\frac{S^n}{K_m^m + S^m}$ — **forward dependence on S** : the enzyme needs substrate to work, and its rate saturates once S is large (Hill-type/Michaelis–Menten-like kinetics).
 - $\frac{1}{1 + k_i P^q}$ — **feedback inhibition**: the product P inhibits the enzyme that produces it. This is the *delayed negative feedback* from the introduction — P must accumulate before it can shut down its own production.

The exponents n , m and q control how sharply (how “switch-like”) S activates, and P inhibits, the enzyme. Throughout this notebook, a_1 — the supply rate of S — is the parameter we vary; it acts as the main control knob (the **bifurcation parameter**) that switches the system between a stable steady state and sustained oscillations.

2.2 Time Series

We start with the most direct test: simulate the ODEs over time for two values of a_1 and look at the trajectories of $S(t)$ and $P(t)$.

- For one value of a_1 , the system should settle down to a constant **steady state**.
- For the other value, S and P should **oscillate indefinitely**, out of phase with each other.

The right-hand panels below show the same trajectories in the S – P plane (the “phase plane” or “state space”): a steady state appears as a single point, while an oscillation appears as a closed loop.

```
[2]: def model(t, variables, a1, b1, a2, b2, k_max, K_m, k_i, n, m, q):
    """
    Right-hand side of the ODE system: returns [dS/dt, dP/dt].

    `variables` packs the two state variables (S, P) into a single
    list, as required by `solve_ivp`. The remaining arguments are the
    fixed model parameters (rate constants and Hill exponents).
    """
    S, P = variables

    # Rate of the enzyme reaction that converts S -> P.
    # - S**n / (K_m**m + S**m): forward dependence on S - saturating,
    #   Hill-type kinetics (more substrate -> faster reaction, up to a limit)
    # - 1 / (1 + k_i * P**q): feedback inhibition - more product P
    #   slows the reaction down (delayed negative feedback)
    enzymatic_rate = (k_max * S**n) / (K_m**m + S**m) / (1 + k_i * P**q)

    # S is produced at constant rate a1, removed by first-order decay
    # (b1*S), and consumed by the enzyme reaction.
    dSdt = a1 - b1 * S - enzymatic_rate
```

```

# P is produced at constant rate a2, removed by first-order decay
# (b2*P), and produced by the enzyme reaction - the flux that
# leaves S arrives in P, so the same term appears with a + sign here.
dPdt = a2 - b2 * P + enzymatic_rate

return [dSdt, dPdt]

# Initial concentrations [S(0), P(0)] - same starting point for both runs below
y0 = [0.5, 6.97]

# We run the SAME model twice, changing only a1 (the supply rate of S).
# a1 is the parameter that determines whether the system settles to a
# steady state or oscillates - see the Bifurcation analysis section later.
a1 = [0.58, 0.62] # a1[0] -> steady state, a1[1] -> sustained oscillation

# Fixed parameters, kept the same for both runs
a2 = 0.02
b1, b2 = 0.18, 0.05

k_max, K_m, k_i = 25.0, 0.7, 0.06
n, m, q = 1, 3, 3

# Time interval to integrate over
t_span = (0, 700)

colors = ['tomato', 'deepskyblue']

# Layout: one row per a1 value (steady state / oscillation),
# left column = concentrations vs time, right column = phase plane (S vs P)
fig, ax = subplots(ncols=2, nrows=2, figsize=(8, 4))

for index, color in enumerate(colors):

    # Numerically integrate the ODEs with solve_ivp.
    # - method='BDF': an implicit solver well suited to "stiff" systems,
    #   where some variables change much faster than others (common in
    #   biochemical models with very different rate constants).
    # - max_step=0.1: forces small time steps so that fast transients
    #   (e.g. the rapid part of a relaxation oscillation) are not missed.
    solution = solve_ivp(model, t_span, y0, args=(a1[index], b1, a2, b2, k_max,
    ↪ K_m, k_i, n, m, q), method='BDF', max_step=0.1)

    t = solution.t

    S = solution.y[0]
    P = solution.y[1]

```

```

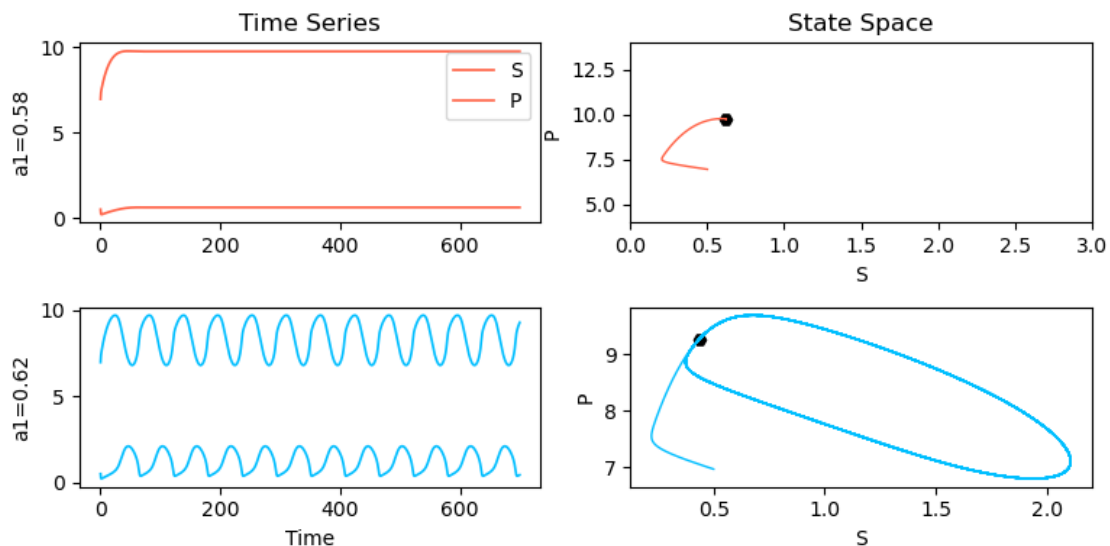
# Left column: concentrations of S and P over time
ax[index, 0].plot(t, S, label='S', linewidth=1.2, color=colors[index])
ax[index, 0].plot(t, P, label='P', linewidth=1.2, color=colors[index])
# Right column: phase-plane trajectory (S vs P); the marker shows
# where the trajectory ends up at t = t_span[1]
ax[index, 1].scatter(S[-1], P[-1], c='k', marker='H')
ax[index, 1].plot(S, P, linewidth=1, color=colors[index]);
ax[index, 0].set_ylabel(f'a1={a1[index]}')

ax[1, 0].set_xlabel('Time')
ax[0, 0].legend()
ax[0, 0].set_title('Time Series')

ax[0, 1].set_xlabel('S')
ax[1, 1].set_xlabel('S')
ax[0, 1].set_ylabel('P')
ax[1, 1].set_ylabel('P')
ax[0, 1].set_title('State Space')
ax[0, 1].set_xlim(0, 3)
ax[0, 1].set_ylim(4, 14)

fig.tight_layout()

```



```

[4]: # Final (S, P) values reached at t = 700 for the SECOND run above (a1 = 0.62),
# rounded to 2 decimals for readability.

```

```
# If that run is oscillating, this is just a snapshot at one point in time -
# NOT a fixed point - the system keeps moving after this.
np.around((S[-1], P[-1]), 2)
```

```
[4]: array([0.44, 9.28])
```

2.3 State Space Nullclines

The time series tell us *what* happens, but not *why*. To understand why the system oscillates for some parameter values and not others, it helps to look at the phase plane more carefully, using two classic tools from dynamical systems theory:

- A **nullcline** is the set of points where one variable's rate of change is zero. The *S*-nullcline is where $dS/dt = 0$; the *P*-nullcline is where $dP/dt = 0$.
- A **fixed point** (steady state) is a point where *both* rates of change are zero at once — i.e. where the two nullclines intersect.

If the fixed point is **stable**, nearby trajectories spiral *into* it (this matches the steady-state behaviour seen above for one value of a_1). If it is **unstable**, nearby trajectories spiral *away* from it — but they cannot run off to infinity, so instead they settle onto a surrounding closed loop: the **limit cycle** corresponding to the oscillation.

The cell below computes both nullclines numerically (root-finding with `fsolve`), locates the fixed point, and overlays a simulated trajectory to reveal the limit cycle directly.

```
[4]: import plotly.graph_objects as go
from scipy.optimize import fsolve

# Oscillatory parameters - we fix a1 at a single value for this analysis
# (one of the parameter sets that produces sustained oscillations;
# compare with the bifurcation analysis later in the notebook)
a1_fixed = 0.65

print("NULLCLINE ANALYSIS FOR OSCILLATORY SYSTEM")
print("="*50)
print(f"b1={b1}, b2={b2}")
print(f"a1={a1_fixed}, a2={a2}")
print(f"k_max={k_max}, K_m={K_m}")
print(f"k_i={k_i}")
print(f"n={n}, m={m}, q={q}")
print()

def enzymatic_rate(S, P):
    """Enzymatic rate with forward and feedback inhibition"""
    return (k_max * S**n) / (K_m**m + S**m) / (1 + k_i * P**q)

def S_nullcline(S, P):
    """S-nullcline: dS/dt = 0 => a1 - b1*S = v(S,P)"""
    return a1_fixed - b1*S - enzymatic_rate(S, P)
```

```

def P_nullcline(S, P):
    """P-nullcline:  $dP/dt = 0 \Rightarrow a_2 - b_2 \cdot P = -v(S, P)$ """
    return a2 - b2*P + enzymatic_rate(S, P)

# --- S-nullcline -----
# For each value of S, find the value(s) of P that satisfy  $dS/dt = 0$ .
# This is a 1D root-finding problem (in P) for each fixed S, solved with
# fsolve. We try several starting guesses for P because the equation can
# have more than one solution (the nullcline can be multi-valued in P).
print("Computing S-nullcline...")
S_values = np.linspace(0.1, 3, 300)
P_S_null = []

for S in S_values:
    def equation(P):
        return S_nullcline(S, P)

    try:
        # Try multiple initial guesses
        solutions = []
        for P_guess in [5.0, 6.0, 7.0, 8.0]:
            result = fsolve(equation, P_guess)
            if (result[0] > 0 and result[0] < 15 and
                abs(equation(result[0])) < 1e-6):
                solutions.append(result[0])

        # Remove duplicates and take the first valid solution
        if solutions:
            unique_solutions = []
            for sol in solutions:
                if not any(abs(sol - existing) < 0.1 for existing in
unique_solutions):
                    unique_solutions.append(sol)
            P_S_null.append(unique_solutions[0]) # Take first branch for
plotting
        else:
            P_S_null.append(np.nan)
    except:
        P_S_null.append(np.nan)

# --- P-nullcline -----
# Same idea as above, but now solving  $dP/dt = 0$  for P at each fixed S.
print("Computing P-nullcline...")
P_P_null = []

for S in S_values:

```

```

def equation(P):
    return P_nullcline(S, P)

try:
    # Try multiple initial guesses
    solutions = []
    for P_guess in [0.1, 1.0, 3.0, 5.0]:
        result = fsolve(equation, P_guess)
        if (result[0] > 0 and result[0] < 15 and
            abs(equation(result[0])) < 1e-6):
            solutions.append(result[0])

    # Remove duplicates and take the first valid solution
    if solutions:
        unique_solutions = []
        for sol in solutions:
            if not any(abs(sol - existing) < 0.1 for existing in
                unique_solutions):
                unique_solutions.append(sol)
        P_P_null.append(unique_solutions[0]) # Take first branch for
    plotting
    else:
        P_P_null.append(np.nan)
    except:
        P_P_null.append(np.nan)

# --- Plot the phase portrait -----
fig = go.Figure()

# Plot S-nullcline
valid_S_S = []
valid_P_S = []
for S, P in zip(S_values, P_S_null):
    if not np.isnan(P):
        valid_S_S.append(S)
        valid_P_S.append(P)

fig.add_trace(go.Scatter(x=valid_S_S, y=valid_P_S, mode='lines',
                        name='S-nullcline (dS/dt=0)',
                        line=dict(color='blue', width=2)))

# Plot P-nullcline
valid_S_P = []
valid_P_P = []
for S, P in zip(S_values, P_P_null):
    if not np.isnan(P):
        valid_S_P.append(S)

```



```

        valid_P_P.append(P)

fig.add_trace(go.Scatter(x=valid_S_P, y=valid_P_P, mode='lines',
                        name='P-nullcline (dP/dt=0)',
                        line=dict(color='red', width=2)))

# --- Fixed point(s): where BOTH nullclines are satisfied simultaneously ---
# Search on a grid of (S, P) starting guesses and solve the 2x2 system
# [S_nullcline, P_nullcline] = [0, 0] with fsolve from each starting point.
# Duplicate solutions (within 0.1 of an existing one) are discarded.
print("Finding fixed points...")
fixed_points = []

# Search in S-P space
S_search = np.linspace(0.01, 1.2, 50)
P_search = np.linspace(9, 11, 50)

for S_guess in S_search:
    for P_guess in P_search:
        def equations(vars):
            S, P = vars
            return [S_nullcline(S, P), P_nullcline(S, P)]

        try:
            result = fsolve(equations, [S_guess, P_guess])
            S_sol, P_sol = result

            # Check if valid solution
            residuals = equations([S_sol, P_sol])
            if (np.linalg.norm(residuals) < 1e-4 and
                S_sol > 0 and P_sol > 0 and
                S_sol <= 2 and P_sol <= 15):

                # Check if new
                is_new = True
                for existing in fixed_points:
                    if (abs(S_sol - existing[0]) < 0.1 and
                        abs(P_sol - existing[1]) < 0.1):
                        is_new = False
                        break

                if is_new:
                    fixed_points.append((S_sol, P_sol))
        except:
            continue

print(f"Found {len(fixed_points)} fixed point(s):")

```

```

for i, (S, P) in enumerate(fixed_points):
    print(f"  FP{i+1}: S = {S:.3f}, P = {P:.3f}")

# Plot fixed points
for i, (S, P) in enumerate(fixed_points):
    fig.add_trace(go.Scatter(x=[S], y=[P], mode='markers',
                             marker=dict(size=12, color='green', symbol='circle',
                                           line=dict(width=1, color='black')),
                             name=f'Fixed Point {i+1}'))

# --- Limit cycle: simulate from an arbitrary starting point and discard ---
# the initial transient. If the fixed point is unstable, the trajectory
# will converge onto the surrounding limit cycle.
print("Adding limit cycle from simulation...")
from scipy.integrate import solve_ivp

def system_dynamics(t, variables):
    S, P = variables
    rate = enzymatic_rate(S, P)
    dSdt = a1_fixed - b1*S - rate
    dPdt = a2          - b2*P + rate
    return [dSdt, dPdt]

# Simulate to get limit cycle
sol = solve_ivp(system_dynamics, (0, 2000), [1.0, 1.0],
                 t_eval=np.linspace(0, 2000, 5000), method='RK45')

# Use only the last part to avoid transients
S_cycle = sol.y[0][-2000:]
P_cycle = sol.y[1][-2000:]

fig.add_trace(go.Scatter(x=S_cycle, y=P_cycle, mode='lines',
                         name='Oscillation',
                         line=dict(color='black', width=4)))

# Update layout
fig.update_layout(
    title="Nullclines and Limit Cycle for Oscillatory System",
    xaxis_title="Substrate Concentration S",
    yaxis_title="Product Concentration P",
    width=700,
    height=600,
    showlegend=True
)

fig.update_xaxes(range=[0, 3])
fig.update_yaxes(range=[6, 12])

```

```

fig.show()

# --- Mathematical summary -----
print("\n" + "="*50)
print("MATHEMATICAL ANALYSIS")
print("="*50)

print("Nullcline equations:")
print("S-nullcline:  $a - b \cdot S = k_{\max} \cdot S / (K_m^3 + S^3) \cdot 1 / (1 + k_i \cdot P^2)$ ")
print("P-nullcline:  $a - b \cdot P = -k_{\max} \cdot S / (K_m^3 + S^3) \cdot 1 / (1 + k_i \cdot P^2)$ ")
print()
print("Rearranged:")
print("S-nullcline:  $P = \sqrt{[(k_{\max} \cdot S / ((a - b \cdot S)(K_m^3 + S^3)) - 1) / k_i]}$ ")
print("P-nullcline:  $P = [a + k_{\max} \cdot S / ((K_m^3 + S^3)(1 + k_i \cdot P^2))] / b$ ")
print()

# A single fixed point that is unstable, together with a bounded phase
# plane, implies (by the Poincare-Bendixson theorem) that trajectories
# must converge onto a periodic orbit - the limit cycle plotted above.
if len(fixed_points) == 1:
    S_fp, P_fp = fixed_points[0]
    print(f"Single fixed point at: S = {S_fp:.3f}, P = {P_fp:.3f}")
    print("This is typical for oscillatory systems - an unstable fixed point")
    print("surrounded by a stable limit cycle!")

    # Check stability roughly by simulating nearby
    sol_test = solve_ivp(system_dynamics, (0, 50), [S_fp + 0.1, P_fp + 0.1],
                        t_eval=[0, 50], method='RK45')
    final_S, final_P = sol_test.y[0, -1], sol_test.y[1, -1]
    distance = np.sqrt((final_S - S_fp)**2 + (final_P - P_fp)**2)

    if distance > 0.5:
        print("Fixed point appears UNSTABLE (spiral source)")
    else:
        print("Fixed point appears STABLE")

print("\nThe purple limit cycle shows sustained oscillations around the fixed_
↪point!")

```

NULLCLINE ANALYSIS FOR OSCILLATORY SYSTEM

```

=====
b1=0.18, b2=0.05
a1=0.65, a2=0.02
k_max=25.0, K_m=0.7
k_i=0.06
n=1, m=3, q=3

```

Computing S-nullcline...

Computing P-nullcline...

Finding fixed points...

Found 1 fixed point(s):

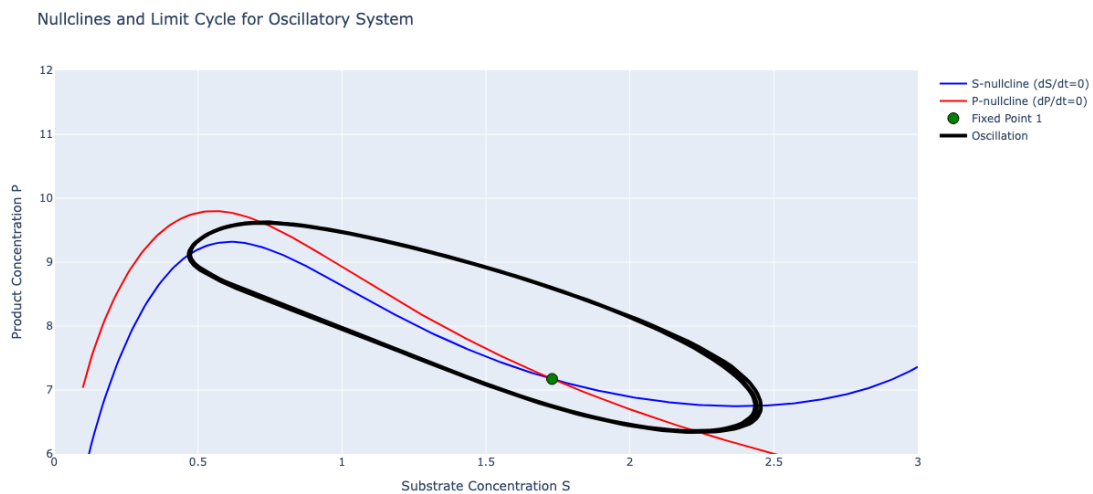
FP1: S = 1.730, P = 7.172

Adding limit cycle from simulation...

/var/folders/cs/lkcj7j890kv6kfxbkl56w9h80000gn/T/ipykernel_23791/2912851335.py:1

29: RuntimeWarning:

The iteration is not making good progress, as measured by the improvement from the last ten iterations.



=====

MATHEMATICAL ANALYSIS

=====

Nullcline equations:

S-nullcline: $a - b \cdot S = k_{\max} \cdot S / (K_m^3 + S^3) \cdot 1 / (1 + k_i \cdot P^2)$

P-nullcline: $a - b \cdot P = -k_{\max} \cdot S / (K_m^3 + S^3) \cdot 1 / (1 + k_i \cdot P^2)$

Rearranged:

S-nullcline: $P = \sqrt{[(k_{\max} \cdot S / ((a - b \cdot S)(K_m^3 + S^3))) - 1] / k_i}$

P-nullcline: $P = [a + k_{\max} \cdot S / ((K_m^3 + S^3)(1 + k_i \cdot P^2))] / b$

Single fixed point at: S = 1.730, P = 7.172

This is typical for oscillatory systems - an unstable fixed point surrounded by a stable limit cycle!

Fixed point appears STABLE

The purple limit cycle shows sustained oscillations around the fixed point!

[]:

2.4 Oscillation animation

Numbers and curves can feel abstract — this final visualisation turns the oscillation into something more intuitive. Two circles represent the concentrations of S and P : as each concentration rises and falls over time, its circle grows and shrinks. Because S is consumed to produce P (and P feeds back to inhibit that conversion), the two circles pulse **out of phase** — as S grows, P shrinks, and vice versa. This alternating “charge / discharge” pattern is the visual signature of the relaxation oscillator described in the introduction.

Use the **Play** button (or drag the time slider) to watch the oscillation unfold.

```
[5]: import plotly.graph_objects as go

# -----
# This cell re-simulates the model for a1 = 0.62 (the oscillatory case from
# the Time Series section above) and then builds an animated plotly figure:
# - one circle for S, one circle for P, whose RADII are rescaled versions
#   of their concentrations over time
# - a green arrow representing the "flow" from S to P
# - a time slider and Play/Pause buttons to step through the simulation
# Most of the code below is plotly bookkeeping (shapes, frames, layout);
# the parts that matter biologically are the model/parameters at the top
# and the radius normalisation just below the simulation.
# -----

# Model definition (identical to the `model` function used earlier)
def model(t, variables, a1, b1, a2, b2, k_max, K_m, k_i, n, m, q):
    """Coupled system with feedback inhibition"""
    S, P = variables

    enzymatic_rate = (k_max * S**n) / (K_m**m + S**m) / (1 + k_i * P**q)

    dSdt = a1 - b1 * S - enzymatic_rate
    dPdt = a2 - b2 * P + enzymatic_rate

    return [dSdt, dPdt]

S_0 = 1.97
P_0 = 6.97

# Parameters
a1, a2 = 0.62, 0.02
b1, b2 = 0.18, 0.05
```

```

k_max, K_m, k_i = 25.0, 0.7, 0.06
n, m, q = 1, 3, 3

t_span = (0, 700)

# Solve the ODE for a1 = 0.62 (oscillatory case)
t_eval = np.linspace(t_span[0], t_span[1], 1000) # Even fewer points for
↳stability
sol = solve_ivp(model, t_span, [S_0, P_0], args=(a1, b1, a2, b2, k_max, K_m,
↳k_i, n, m, q),
                t_eval=t_eval, method='RK45')

S_sol = sol.y[0]
P_sol = sol.y[1]
t_sol = sol.t

# Rescale S and P to a fixed radius range [min_radius, max_radius],
# so each species is drawn as a circle that visibly grows/shrinks
# over the course of the oscillation, regardless of its absolute
# concentration units.
# Normalize for circle radii - using pixel units for shapes
min_radius, max_radius = 15, 40
S_radius = min_radius + (max_radius - min_radius) * (S_sol - np.min(S_sol)) /
↳(np.max(S_sol) - np.min(S_sol))
P_radius = min_radius + (max_radius - min_radius) * (P_sol - np.min(P_sol)) /
↳(np.max(P_sol) - np.min(P_sol))

# Fixed positions
S_pos = [-4, 0]
P_pos = [4, 0]

# Build the animation: one "frame" per simulated time point, each
# redrawing the S and P circles (and the connecting arrow) at the
# size corresponding to that time point's concentrations.
# Create the figure with shapes (much more stable than scatter plots)
fig = go.Figure()

# Create frames using shapes instead of scatter plots
frames = []
for i in range(len(t_sol)):
    frame_data = [
        # S circle as shape
        dict(
            type="circle",
            xref="x", yref="y",
            x0=S_pos[0] - S_radius[i]/20, # Divide by scaling factor

```

```

        y0=S_pos[1] - S_radius[i]/20,
        x1=S_pos[0] + S_radius[i]/20,
        y1=S_pos[1] + S_radius[i]/20,
        line_color="blue",
        fillcolor="blue",
        line_width=2,
        opacity=0.7
    ),
    # P circle as shape
    dict(
        type="circle",
        xref="x", yref="y",
        x0=P_pos[0] - P_radius[i]/20,
        y0=P_pos[1] - P_radius[i]/20,
        x1=P_pos[0] + P_radius[i]/20,
        y1=P_pos[1] + P_radius[i]/20,
        line_color="red",
        fillcolor="red",
        line_width=2,
        opacity=0.7
    ),
    # Arrow line
    dict(
        type="line",
        xref="x", yref="y",
        x0=S_pos[0] + S_radius[i]/20,
        y0=S_pos[1],
        x1=P_pos[0] - P_radius[i]/20,
        y1=P_pos[1],
        line=dict(color="green", width=4)
    ),
    # Arrow head
    dict(
        type="path",
        xref="x", yref="y",
        path=f"M {P_pos[0] - P_radius[i]/20 - 0.3} {P_pos[1] - 0.2} L_
↪{P_pos[0] - P_radius[i]/20} {P_pos[1]} L {P_pos[0] - P_radius[i]/20 - 0.3}_
↪{P_pos[1] + 0.2} Z",
        line_color="green",
        fillcolor="green"
    )
]

frame = go.Frame(
    data=[], # No scatter data
    layout=dict(
        shapes=frame_data,

```

```

        # annotations=text_annotations
    ),
    name=f'frame_{i}'
)
frames.append(frame)

# Add initial shapes
fig.add_shape(
    type="circle",
    xref="x", yref="y",
    x0=S_pos[0] - S_radius[0]/20,
    y0=S_pos[1] - S_radius[0]/20,
    x1=S_pos[0] + S_radius[0]/20,
    y1=S_pos[1] + S_radius[0]/20,
    line_color="blue",
    fillcolor="blue",
    line_width=2,
    opacity=0.7
)

fig.add_shape(
    type="circle",
    xref="x", yref="y",
    x0=P_pos[0] - P_radius[0]/20,
    y0=P_pos[1] - P_radius[0]/20,
    x1=P_pos[0] + P_radius[0]/20,
    y1=P_pos[1] + P_radius[0]/20,
    line_color="red",
    fillcolor="red",
    line_width=2,
    opacity=0.7
)

fig.add_shape(
    type="line",
    xref="x", yref="y",
    x0=S_pos[0] + S_radius[0]/20,
    y0=S_pos[1],
    x1=P_pos[0] - P_radius[0]/20,
    y1=P_pos[1],
    line=dict(color="green", width=4)
)

fig.add_shape(
    type="path",
    xref="x", yref="y",

```



```

    path=f"M {P_pos[0] - P_radius[0]/20 - 0.3} {P_pos[1] - 0.2} L {P_pos[0] - P_radius[0]/20} {P_pos[1]} L {P_pos[0] - P_radius[0]/20 - 0.3} {P_pos[1] + 0.2} Z",
    line_color="green",
    fillcolor="green"
)

# Add initial annotations
fig.add_annotation(
    x=S_pos[0],
    y=S_pos[1] - max(S_radius[0], P_radius[0])/15 - 0.8,
    # text=f"S: {S_sol[0]:.2f}",
    showarrow=False,
    font=dict(color="blue", size=12),
    xref="x",
    yref="y"
)

fig.add_annotation(
    x=P_pos[0],
    y=P_pos[1] - max(S_radius[0], P_radius[0])/15 - 0.8,
    # text=f"P: {P_sol[0]:.2f}",
    showarrow=False,
    font=dict(color="red", size=12),
    xref="x",
    yref="y"
)

# Configure animation
fig.frames = frames

# Animation controls: a time slider plus Play/Pause buttons
sliders = [dict(
    steps=[dict(
        method="animate",
        args=[f'frame_{k}',
              dict(mode="immediate", frame=dict(duration=50, redraw=True),
                    transition=dict(duration=0))],
        label=f'{t_sol[k]:.1f}'
    ) for k in range(0, len(t_sol), 15)],
    active=0,
    currentvalue=dict(prefix="Time: ", visible=True),
    len=0.9,
    x=0.1,
    y=0
)]

```

```

updatemenus = [dict(
    type="buttons",
    showactive=False,
    buttons=[
        dict(label="Play", method="animate",
            args=[None, dict(frame=dict(duration=50, redraw=True),
                fromcurrent=True, mode="immediate")]),
        dict(label="Pause", method="animate",
            args=[[None], dict(frame=dict(duration=0, redraw=False),
                mode="immediate")])
    ],
    x=0.1, y=-0.15,
    xanchor="right",
    yanchor="top"
)]

# Update layout with FIXED ranges and no auto-scaling
fig.update_layout(
    title="Oscillating Reaction System",
    xaxis=dict(
        range=[-6, 6], # Fixed range
        showticklabels=False,
        showgrid=True,
        gridcolor="lightgray",
        zeroline=False,
        fixedrange=True, # Prevent any scaling
        constrain="domain"
    ),
    yaxis=dict(
        range=[-3, 3], # Fixed range
        showticklabels=False,
        showgrid=True,
        gridcolor="lightgray",
        zeroline=False,
        fixedrange=True, # Prevent any scaling
        scaleanchor="x",
        scaleratio=1
    ),
    plot_bgcolor="white",
    width=800,
    height=500,
    showlegend=False,
    sliders=sliders,
    updatemenus=updatemenus,
    # Critical: disable all auto-scaling and layout recalculations
    autosize=False,
    margin=dict(l=50, r=50, t=80, b=80)
)

```

```

)

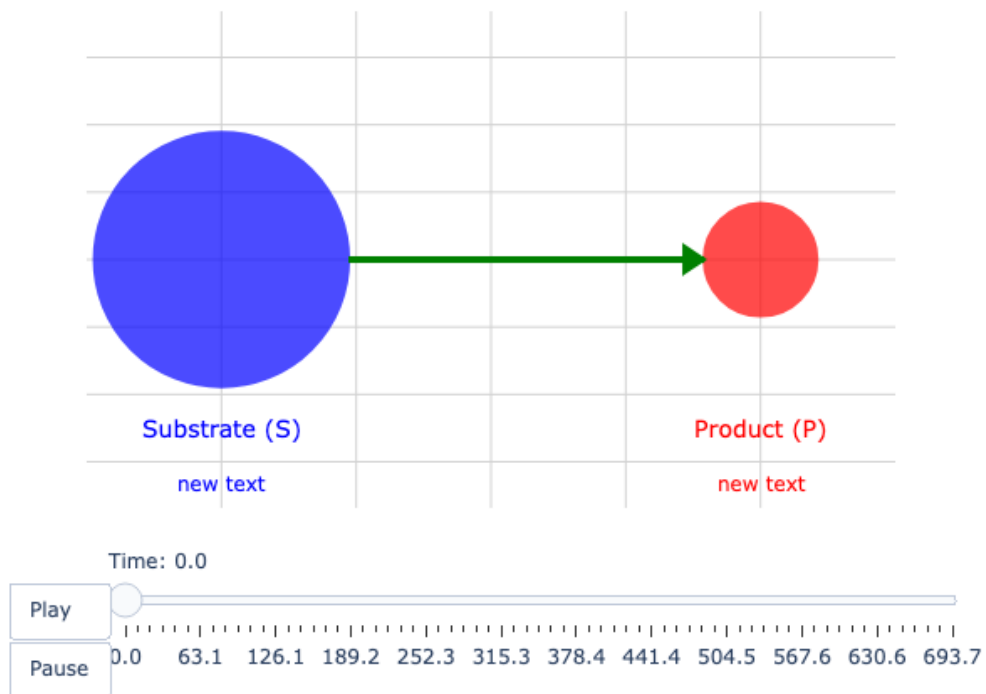
# Add static labels
fig.add_annotation(
    x=S_pos[0], y=-2.5,
    text="Substrate (S)",
    showarrow=False,
    font=dict(color="blue", size=14),
    xref="x",
    yref="y"
)

fig.add_annotation(
    x=P_pos[0], y=-2.5,
    text="Product (P)",
    showarrow=False,
    font=dict(color="red", size=14),
    xref="x",
    yref="y"
)

fig.show()

```

Oscillating Reaction System



[]:

2.5 Bifurcation analysis: how does behaviour change with a_1 ?

The plots above show two specific outcomes (steady state vs. oscillation) for two specific values of a_1 . A natural next question is: **for which values of a_1 does the system oscillate, and how does the oscillation's amplitude change as a_1 varies?**

Answering this systematically means scanning a_1 over a range and, for each value, running the simulation long enough for any transient to die away, then recording the resulting minimum and maximum of S and P :

- If the system settles to a steady state, the minimum and maximum coincide (amplitude = 0).
- If the system oscillates, the minimum and maximum differ, and their difference is the oscillation's amplitude.

A plot of these minima and maxima against a_1 is a **bifurcation diagram**. Points where the behaviour changes qualitatively — where the amplitude becomes non-zero, or returns to zero — are **bifurcation points**. Here this corresponds to a **Hopf bifurcation**: a stable steady state loses stability and a limit cycle (oscillation) is born, or the limit cycle shrinks back down and the steady state regains stability.

2.5.1 The companion script

This scan is implemented in `enzym_oscillations_bifurcation.py`, which lives alongside this notebook. It uses the same model equations as above (with the feedback-inhibition exponent set to $q = 2.8$ rather than $q = 3$), and for each of 150 values of a_1 between 0.63 and 0.72 it:

1. Integrates the ODEs for a long “transient” period (500 time units) plus an “analysis” window (200 time units), using a stiff solver (LSODA).
2. Records the minimum and maximum of S and P over the analysis window.
3. Uses the final state as the initial condition for the next a_1 value — a simple form of *continuation* that helps the solver stay on the same solution branch as a_1 changes.
4. Plots the resulting minima/maxima against a_1 and prints a short text summary of where oscillations occur.

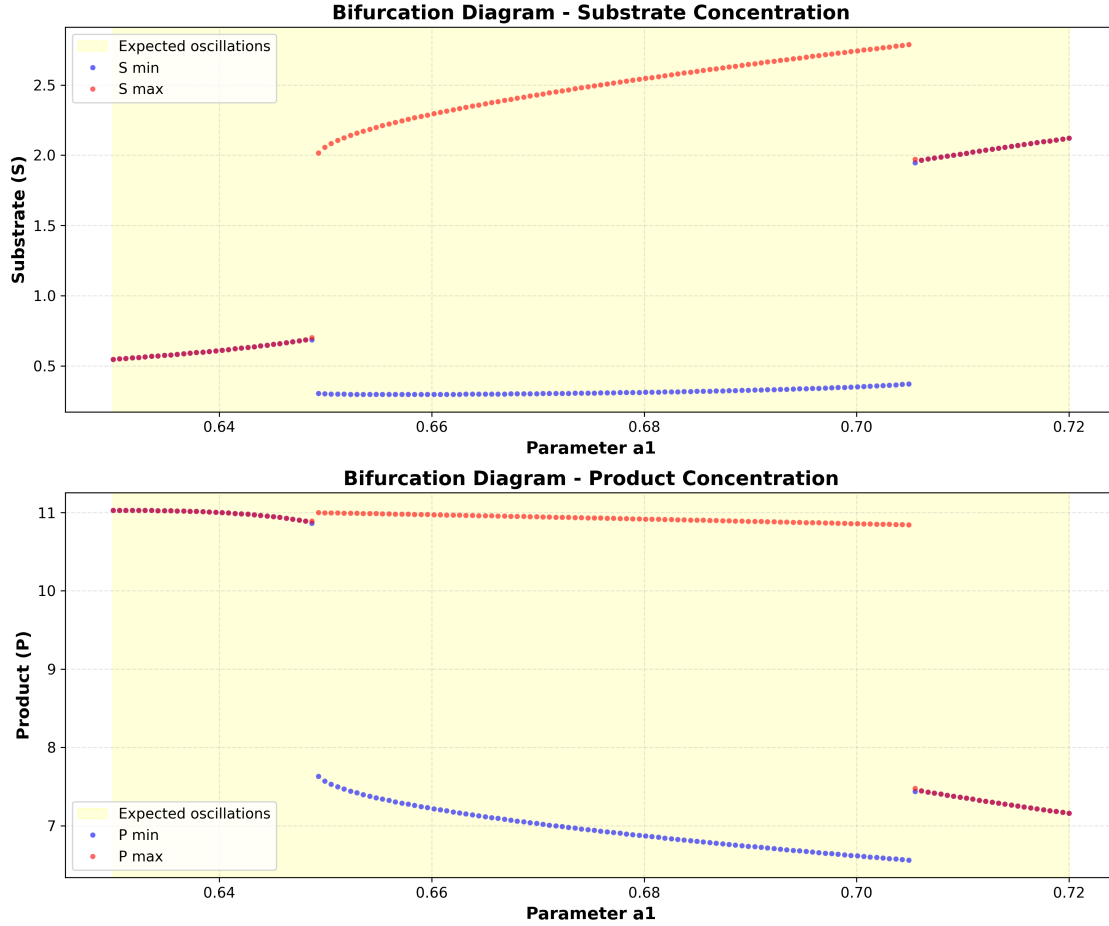
Run it from a terminal with:

```
python enzym_oscillations_bifurcation.py
```

It saves a figure (`bifurcation_diagram_oscillations.png`) and prints a summary to the terminal. (The script is kept separate from this notebook because a full bifurcation scan — many long simulations in a row — can take a while to run.)

2.5.2 Results

Running the script over $a_1 \in [0.63, 0.72]$ gives the bifurcation diagram below:



The printed summary reports:

- Oscillations occur for $a_1 \in [0.6487, 0.7055]$ (95 of the 150 scanned points). Outside this window, S and P sit at a constant steady state (minimum = maximum, i.e. amplitude = 0).
- The largest oscillation amplitude is reached near the *upper* end of this window: amplitude of $S \approx 2.42$ and amplitude of $P \approx 4.28$ at $a_1 \approx 0.705$.
- Just inside either edge of the window the amplitude shrinks towards zero — the signature of a Hopf bifurcation, where a small-amplitude limit cycle is born (or dies) around the fixed point.

A note on parameters. The script uses $q = 2.8$, slightly different from the $q = 3$ used in the cells above. This shifts the oscillatory window: with $q = 3$ (as used for the Time Series, Nullclines and Animation sections), a quick scan shows oscillations roughly for $a_1 \in (0.59, 0.655)$ — consistent with $a_1 = 0.58$ giving a steady state and $a_1 = 0.62$ and $a_1 = 0.65$ both giving oscillations, exactly as seen earlier in the notebook. The general lesson does not depend on the exact value of q : there is always a *finite window* of a_1 bounded by two Hopf bifurcations, and changing other parameters (like q) shifts where that window sits. Try editing q (or k_i , K_m , ...) in the script and re-running it to see this for yourself.

2.6 Summary: what can we learn from these simulations?

A handful of ingredients is enough to oscillate. The model has only two state variables (S , P) and a small number of parameters, yet it can settle to a steady state *or* oscillate indefinitely, purely as a consequence of its chemical kinetics — no external clock or rhythmic input is needed. This is the essence of self-organised behaviour in biochemical networks.

The nullclines explain the mechanism. Forward dependence on S together with feedback inhibition by P shapes the nullclines so that, for some parameter values, they intersect at a single, *unstable* fixed point. Because trajectories cannot escape to infinity (concentrations stay bounded), an unstable fixed point in a bounded 2D phase plane forces trajectories onto a surrounding closed orbit — the limit cycle we see as a sustained oscillation.

a_1 acts as a switch between dynamical regimes. The bifurcation analysis shows that oscillations occur only within a *finite window* of a_1 , bounded by two Hopf bifurcations. Outside this window the system is at a stable steady state; inside it, the system oscillates, with the amplitude growing from zero near each bifurcation point. Biologically, a_1 represents a supply rate — so a cell could move into or out of an oscillatory regime simply by up- or down-regulating the production of S (e.g. via an upstream signalling pathway), without needing to rebuild the whole network.

The animation makes the relaxation-oscillator pattern visible. S and P rise and fall out of phase, with periods of slow change punctuated by faster swings — the qualitative pattern shared by many real biological oscillators (e.g. the calcium spikes of Woods et al., 1986).

This is a minimal example of a recurring motif. The combination of production, removal, and delayed negative feedback recurs throughout cell biology, in systems that look very different at the molecular level: calcium signalling, glycolytic oscillations, the cyclin-CDK cell-cycle oscillator, and circadian clocks (Tyson, Chen & Novak, 2003; Fall et al., 2002). Recognising this shared “dynamical vocabulary” — feedback loops, nullclines, fixed points, bifurcations, limit cycles — is often more useful for understanding *why* a system behaves as it does than memorising the specific molecular players.

2.6.1 Ideas for further exploration

- Change k_i , K_m , n , m or q in the Time Series cell and re-run — how does the oscillatory window (found via the bifurcation script) shift?
- Pick an a_1 value just outside the oscillatory window and run the nullcline cell — how do the nullclines and fixed point change compared with $a_1 = 0.65$?
- Try a longer or shorter t_{span} in the Time Series cell for an oscillatory a_1 — how many cycles does it take to reach the limit cycle from the initial condition?
- Modify the bifurcation script to scan a *different* parameter (e.g. k_i) instead of a_1 , holding a_1 fixed — does the same qualitative picture (a finite oscillatory window) appear?

2.7 References and further reading

2.7.1 Papers and book provided with this notebook

- Woods, N.M., Cuthbertson, K.S.R. & Cobbold, P.H. (1986). *Repetitive transient rises in cytoplasmic free calcium in hormone-stimulated hepatocytes*. Nature 319, 600-602. (CalciumOscillations Woods1986.pdf) — the classic experimental demonstration that hormone signalling produces repetitive calcium *spikes* rather than a sustained rise, with frequency

encoding signal strength. A great example of why the kind of dynamics studied in this notebook matters physiologically.

- Tyson, J.J., Chen, K.C. & Novak, B. (2003). *Sniffers, buzzers, toggles and blinkers: dynamics of regulatory and signaling pathways in the cell*. Current Opinion in Cell Biology 15, 221-231. (Sniffers Tyson2003.pdf) — a short, readable review introducing the “dynamical vocabulary” (feedback, switches, oscillators) used throughout this notebook, with many biological examples.
- Fall, C.P., Marland, E.S., Wagner, J.M. & Tyson, J.J. (eds.) (2002). *Computational Cell Biology*. Springer (Interdisciplinary Applied Mathematics, vol. 20). (Computational Cell Biology FallTyson2000.pdf) — a textbook aimed at biology and mathematics students together. Chapter 2 (on voltage-gated ionic currents and the Morris-Lecar model) contains an especially clear, general discussion of *why* oscillations occur in dynamical systems, including nullclines, stability and the Poincare-Bendixson argument used informally above.

2.7.2 Beginner-friendly further reading (not included as files)

- Strogatz, S.H. *Nonlinear Dynamics and Chaos* (2nd ed., 2015), Westview Press. The standard, very accessible introduction to phase planes, nullclines, fixed points, limit cycles and bifurcations — written for readers without a strong mathematics background.
- Novak, B. & Tyson, J.J. (2008). *Design principles of biochemical oscillators*. Nature Reviews Molecular Cell Biology 9, 981-991. A review connecting the simple oscillator “design principles” introduced here to real cell-cycle and circadian oscillators.
- Ferrell, J.E., Tsai, T.Y.-C. & Yang, Q. (2011). *Modeling the cell cycle: why do certain circuits oscillate?* Cell 144, 874-885. A particularly approachable explanation of how feedback and delay combine to produce oscillations, using the cell cycle as the running example.
- Goldbeter, A. *Biochemical Oscillations and Cellular Rhythms* (1996), Cambridge University Press. A comprehensive but accessible survey of biological oscillators, from glycolysis to circadian rhythms, by one of the field’s pioneers.